- → Data Analyst [SQL, Tableau / PowerBI, Python, Pandas, Excel]
- → Applied Scientist [✓]
- ✓ → ML Engineer [ ]
- → Data Scientist ( ] ✓
- → MLOps

→ Break until 22:20 PM

→ Performance of Rec Sys

1) HitRatio@K

K=5

→ top-5

P1, P4, P6, P8, P9

→ User Watched (Actuals)

P1, P10, P20, P4, P8, P12, P16

$$\text{Recall}@5 = \frac{TP}{TP + FN} = \frac{TP}{\text{No of Actuals}} = \frac{3}{7}$$

Hit Ratio @5

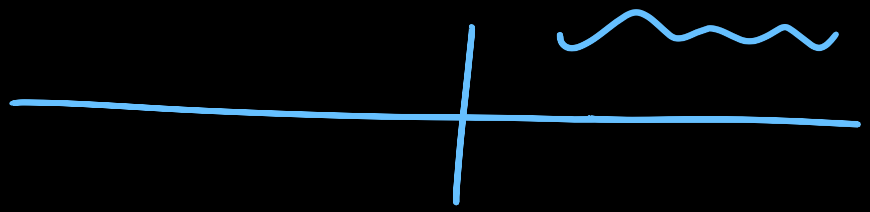
→ top-5

P1, P4, P6, P8, P9 → $\frac{3}{7}$

→ top-5

P1, P4, P8, P6, P9 → $\frac{3}{7}$

AP



21-07-2025

→ top-5 (Recomm)

P1, P4, P6, P8, P9

→ User Watched (Actuals)

P1, P10, P20, P4, P8, P12, P16

$$\text{Precision@5} = \frac{TP}{TP+FP} = \frac{TP}{\text{No of Rec.}} = \frac{3}{5}$$

$$\underset{K=1}{P@1} = \frac{1}{1} \qquad P@2 = \frac{2}{2} \qquad P@3 = \frac{2}{3}$$

$$\underset{K=4}{P@4} = \frac{3}{4}$$

TP = Pink

@1 $\frac{1}{1}$

@2 $\frac{1}{2}$

@3 $\frac{1}{3}$

@4 $\frac{2}{4}$

@5 $\frac{3}{5}$

Rank	Item	Precision@k
1	✓	1/1 = 1 × 1
2		1/2 = 0.5 × 0
3		1/3 = 0.33 × 0
4	✓	2/4 = 0.5 × 1
5	✓	3/5 = 0.6 × 1
6		3/6 = 0.5 × 0

Not relevant Relevant

Average precision

$$AP@6 = \frac{1 + 0.5 + 0.6}{3} = 0.7$$

TP = Pink Color

AP → 0 to 1

1 ← TP

2/2

3/3

	Precision @k
1	1
2	1
3	1
4	0.75
5	0.6
6	0.5

AP = 3/3 = 1

		Precision @k
1	1/1	1
2	2/2	1
3		0.67
4		0.5
5		0.4
6	3/6	0.5

AP = 2.5/3 = 0.83

	Precision @k
1	0
2	0
3	0
4	0.25
5	0.4
6	0.5

AP = 1.15/3 = 0.38

→ P@4 = $\frac{1}{4}$

→ P@5 = $\frac{2}{5}$ = 0.4

→ P@5 = $\frac{3}{6}$ = 0.5